

Solution Requirements

Phase: Requirement Analysis

Project: ShopEEZ | Date: June 2026 | Stack: MongoDB · Express · React · Node.js

1. Functional Requirements — Authentication

REQ#	Requirement	API / Implementation
FR-01	Allow users to register with a unique email and password	POST /api/auth/register
FR-02	Authenticate users and return a signed JWT token	POST /api/auth/login
FR-03	Hash all passwords using Bcrypt before database storage	User.js pre-save hook
FR-04	Expose protected endpoint to fetch authenticated user profile	GET /api/auth/me
FR-05	Allow users to update profile (name, address, phone)	PUT /api/auth/update-profile
FR-06	Allow users to change password with current password verification	PUT /api/auth/change-password
FR-07	All admin routes require valid JWT and role = admin	protect + admin middleware

2. Functional Requirements — Products

REQ#	Requirement	API Endpoint
FR-08	Paginated catalog with search, category, price, rating, sort filters	GET /api/products
FR-09	Return single product full details by ID	GET /api/products/:id
FR-10	Return all product categories dynamically from database	GET /api/products/categories
FR-11	Return curated list of featured products	GET /api/products/featured
FR-12	Admin: add new product with up to 5 images via Multer	POST /api/admin/products
FR-13	Admin: update product details and images	PUT /api/admin/products/:id
FR-14	Admin: delete product	DELETE /api/admin/products/:id

3. Functional Requirements — Orders & Admin

REQ#	Requirement	API Endpoint
FR-15	Customer: place order with shipping address and payment method	POST /api/orders
FR-16	Customer: view all past orders	GET /api/orders/my-orders
FR-17	Admin: see all orders in system with customer details	GET /api/admin/orders
FR-18	Admin: update order status	PUT /api/admin/orders/:id/status
FR-19	Dashboard: display total products, orders, users, revenue	GET /api/admin/dashboard
FR-20	Dashboard: display low stock alerts (stock <= 5)	GET /api/admin/dashboard
FR-21	Admin: view and delete non-admin user accounts	DELETE /api/admin/users/:id

4. Non-Functional Requirements

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API responses complete within 500ms under normal load

NFR-02	Security	JWT tokens stored in localStorage, cleared on logout
NFR-03	Security	All passwords hashed with Bcrypt (salt rounds: 10)
NFR-04	Security	Admin endpoints reject non-admin requests with HTTP 403
NFR-05	Usability	UI fully responsive: mobile 320px, tablet 768px, desktop 1440px
NFR-06	Reliability	MongoDB Atlas provides automated backups and 99.9% uptime SLA
NFR-07	Maintainability	Backend follows MVC: controllers, models, routes, middleware
NFR-08	Scalability	Product listing supports pagination (10 items per page)
NFR-09	Accessibility	Semantic HTML5 elements and ARIA labels for screen readers